TUGAS K32 FI1101 FISIKA
TUGAS 01

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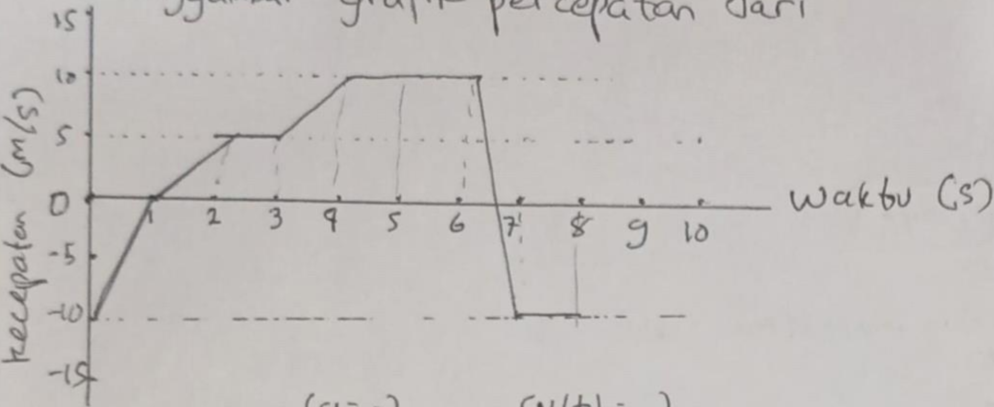
Jawaban

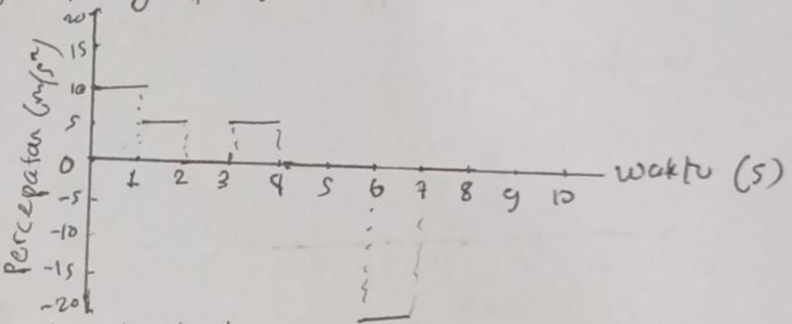
3. Menggambar grafik percepatan dari (kurs)



Menghitung gradien (a=...) dan persamaan (v(t) = ...) saat, maka grafiknya

a.

t	a	$v(t)$
0 — 1	$a = 10$	$v(t) = -10 + 10t$
1 — 2	5	$= -5 + 5t$
2 — 3	0	$= 5$
3 — 4	5	$= -10 + 5t$
4 — 5	0	$= 10$
5 — 6	0	$= 10$
6 — 7	-20	$= 130 - 20t$
7 — 8	0	$= -10$



b. Benda saat $t = 8$ s bisa menggunakan integral; jika $x(0) = 2$ m → menghitung luas bertanda

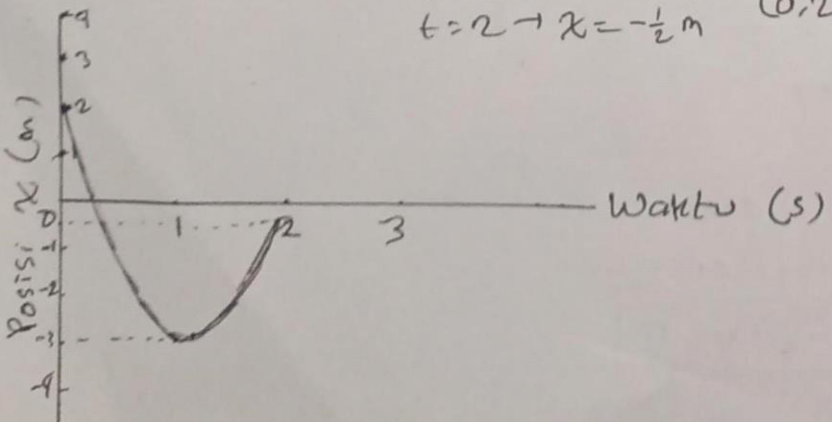
$$x(8) = 2 + \int_0^8 v(t)\,dt$$

$$= 2 + \left(-\frac{10}{2} + \frac{5}{2} + 5 + \left(5 + \frac{5}{2}\right) + 10 + 10 + 0 - 10\right)$$

$$= 2 + 20$$

$$= \boxed{22 \text{ m}}$$

c. Hitung posisi saat $t = 0 \rightarrow x = 2$ m didapat titik-titik
$t = 1 \rightarrow x = -3$ m $(0,2)$; $(1,-3)$; dan $(2,-\frac{1}{2})$
$t = 2 \rightarrow x = -\frac{1}{2}$ m



7. $\vec{r} = 3t\hat{i} + 4t^2\hat{j} + 2\hat{k}$

a. Menentukan $\vec{v}(t)$

$\vec{v}(t) = \frac{d\vec{r}(t)}{dt} = \frac{d}{dt}\left(3t\hat{i} + 4t^2\hat{j} + 2\hat{k}\right)$

$= 3\hat{i} + 8t\hat{j} + 0\hat{k}$

$= \boxed{3\hat{i} + 8t\hat{j}}$ m/s

b. $\vec{v}$ dan $\|\vec{v}\|$ saat $t = 2$

$\vec{v}(t)|2 = \vec{v}(2)$

$= 3\hat{i} + 8(2)\hat{j}$

$= \boxed{3\hat{i} + 16\hat{j} \text{ m/s}}$

$\|\vec{v}(2)\| = \sqrt{3^2 + 16^2}$

$= \sqrt{9 + 256}$

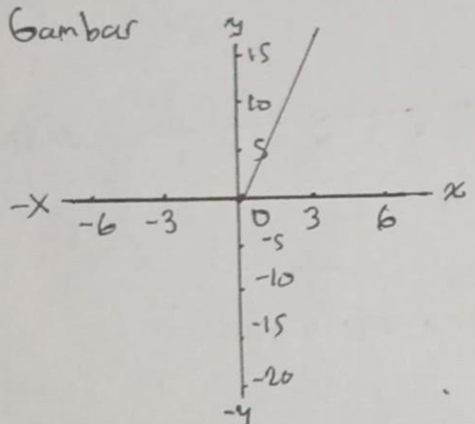
$= \sqrt{265} \approx \boxed{16{,}27882 \text{ m/s}}$

c. Menggambarkan vektor dan menentukan sudut

$\arg(\vec{v}(2)) = \tan^{-1}\left(\frac{\vec{v}_y(2)}{\vec{v}_x(2)}\right)$

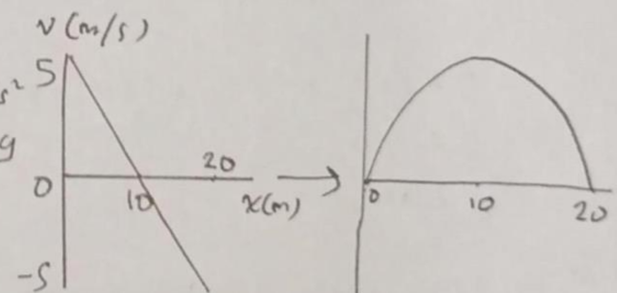
$= \tan^{-1}\left(\frac{16}{3}\right) \approx \boxed{79{,}3803^\circ}$

Gambar



(Referensi pakai soal kuis sehingga ambil $g = 10 \text{ m/s}^2$)

10. $O(0,0)$ dan v_0, θ_0 dengan grafik di samping



a. $v_{0y} = v_{0y}(0) = \boxed{5 \text{ m/s}}$

$v_{0x} = v_x = \frac{20}{t} = \boxed{20 \text{ m/s}}$

$-a = a - gt$

$-5 = 5 - 10t$

$t = 1$ m/s

b. Sudut kemiringan awal $= \theta_0$

$= \tan^{-1}\left(\frac{v_{0y}}{v_{0x}}\right)$

$= \tan^{-1}\left(\frac{5}{20}\right) = \tan^{-1}\left(\frac{1}{4}\right) \approx \boxed{14{,}0362^\circ}$

c. Koordinat tertinggi, yaitu saat $x = 10$ dan $t = \frac{1}{2}$

$s = v_{0y}t - \frac{1}{2}gt^2$

$= 5\left(\frac{1}{2}\right) - \frac{1}{2}\cdot 10\left(\frac{1}{2}\right)^2$

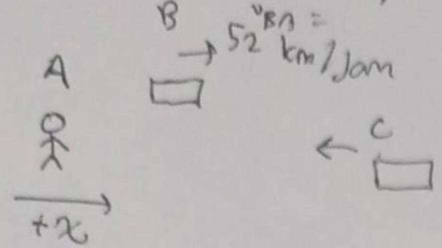
$= \frac{5}{2} - \frac{5}{4}$

$= \frac{5}{4}$

Maka, koordinat tertinggi

$\boxed{h_{max} = \left(10, \frac{5}{4}\right) \text{ meter}}$

1. Soal gerak relatif



a. $V_{BA} = 52$ km/Jam
$V_{CA} = -78$ km/Jam
$V_{CB} = V_{CA} - V_{BA}$
$= -78 - 52$
$= \boxed{-130 \text{ km/Jam}}$ atau $\boxed{\vec{V}_{CB} = (-130\hat{\imath} + 0\hat{\jmath}) \text{ km/Jam}}$

$\rightarrow \boxed{\vec{V}_{CB} = (-81{,}25\hat{\imath} + 0\hat{\jmath}) \text{ m/s}}$

Artinya, kecepatan mobil C menurut B lebih cepat karena berlawanan arah

b. $V_{CAf} = V_{CA} \pm gt$
$0 = -21{,}6\bar{6} + 10 \cdot g$
$\Rightarrow g = +2{,}16\bar{6}$ m/s

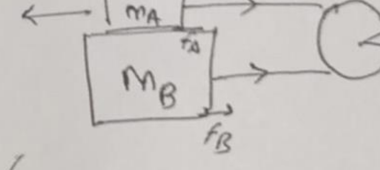
Ubah dari km/jam ke m/s
-78 km/Jam $= -21{,}6\bar{6}$ m/s

Maka, $V_{CA}(t) = -21{,}6\bar{6} + 2{,}16\bar{6}t$ m/s
untuk $0 \le t \le 10$

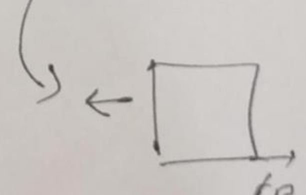
Untuk menghitung kecepatan rata-rata, integralkan dan bagi dengan 10
$= \frac{1}{10} \int_0^{10} V_{CA} \, dt$
$= \frac{1}{10} \left[\frac{2{,}16\bar{6}t^2}{2} - 21{,}6\bar{6}t \right]_0^{10}$
$= \frac{1}{10} (-50 \cdot 2{,}16\bar{6})$
$= -10{,}83\bar{3}$ m/s

Dengan demikian, kecepatan rata-ratanya adalah
$\vec{V}_{CA} = \boxed{(-10{,}83\bar{3}\hat{\imath} + 0\hat{\jmath}) \text{ m/s}}$ atau dalam km/jam, $\vec{V}_{CA} = \boxed{(-39\hat{\imath} + 0\hat{\jmath}) \text{ m/s}}$

3. (Soal kuis)
$m_A = 2$ kg, $m_B = 8$ kg; $\mu_{BS} = 0{,}5$, $\mu_{Bk} = 0{,}3$
$\mu_{AS} = 0{,}4$, $\mu_{Ak} = 0{,}2$

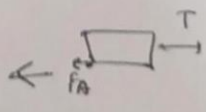


Karena ingin bergerak, $a = 0$ dan menggunakan gaya gesek statis sehingga
$F - f_B - f_A - T = 0 \Rightarrow F = f_A + f_B + T$



$T = f_A$ karena ingin bergerak sehingga

dan



$F = f_A + f_A + f_B$
$= \mu_{AS} \cdot N_A + \mu_{AS} \cdot N_A + \mu_{BS} \cdot N_B$
$= 2\mu_{AS} \cdot m_A \cdot g + \mu_{BS} \cdot (m_A + m_B) g$
$= 2 \cdot 0{,}4 \cdot 2 \cdot 10 + 0{,}5 \cdot 10 \cdot 10$
$= 16 + 50$
$= \boxed{66 \text{ N}}$